

AYUSH PATEL

AI Engineer — AI Core Team, Samsung Research HQ · President, IIT Alumni Association of Korea (IITAAK)

B.Tech, Computer Science and Engineering
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SUMMARY

AI engineer at Samsung Research (AI Core Team) building core AI technology behind Samsung products used worldwide, with six years spanning applied AI, cloud-native platforms and Kubernetes networking at carrier scale. IIT Delhi Computer Science graduate; CKA & LFCS certified; open-source contributor to the cloud-native ecosystem; President of the IIT Alumni Association of Korea, leading the India–Korea technology community.

ACADEMIC DETAILS

Year	Degree	Institute	CGPA/Percentage
2016-2020	B.Tech in Computer Science and Engineering	Indian Institute of Technology Delhi	9.040/10.0
2016	Class XII, CBSE	O. P. Jindal School, Raigarh (C.G.)	97.8%
2014	Class X, CBSE	O. P. Jindal School, Raigarh (C.G.)	10/10

SCHOLASTIC ACHIEVEMENTS

- Secured **All India Rank 192** in JEE Mains and **All India Rank 575** in JEE Advanced among 2 million students
- State Topper** with **97.8%** — **top 0.1%** among 1.06 million candidates in AISSCE Class XII, 2016 (CBSE)
- Selected as **KVPY Scholar** under ‘Kishore Vaigyanik Protsahan Yojana’ organised by Indian Institute of Science
- Awarded **IIT Delhi Merit Scholarship** for being in the **top 7%** among 850 students in the 1st and 2nd semester

CORPORATE EXPERIENCE/ INTERNSHIPS

AI Core Team

Software Engineer (Artificial Intelligence R&D)

Samsung Research (HQ), South Korea

Dec 2025 - Present

- Build core AI technology powering intelligent experiences across Samsung products and services worldwide
- Take AI research from prototype to production in collaboration with research labs and business units across Samsung
- Work across the AI stack — models, data, platforms and services — delivering AI features at global product scale

SmartHome AI Lab

Software Engineer (Artificial Intelligence R&D)

Samsung Research (HQ), South Korea

Jul 2024 - Dec 2025

- Built AI experiences for millions of connected homes across the SmartThings ecosystem
- Developed AI-driven home intelligence connecting devices, users and services across the smart home

Strategic Partnership Management & Business Strategy

Solution Architect and Strategic Partnership/ Business Strategy

Samsung Networks (HQ), South Korea

Nov 2023 - Jul 2024

- Design and manage the System Integration (SI) Solution Architecture of 5G/O-RAN for various COTS, CaaS, RU/MMU
- Layout process for System Integration and service delivery of Service Management and Orchestration (SMO) and rApps
- Strategic Partnership Management (new business strategies, maintain relation, etc.) for Hyperscalers and COTS vendors

Advanced Cloud Native Solutions

Software R&D (Cloud Platform/Infrastructure Development & Operations)

Samsung Networks (HQ), South Korea

Oct 2020 - Oct 2023

- Develop Cloud Native Load Balancer for various protocols for 5G vCore & vRAN using Kubernetes Operator Framework
- Work with cloud native tools & plugins like helm, ansible, multus, cilium, DPDK, SRIOV etc. on public & private cloud
- Contribute to open-source projects like whereabouts, multus and other libraries and participate in community discussions

Tech Lead

Co-founder (Management & Operations)

Halanx Technologies Pvt. Ltd.

Jan 2020 - May 2020

- Led the Technical team comprising of App Developers, Backend Developers, Frontend Developers and UI/UX Designers

- Developed Rental Housing Solution to reduce the operating cost of owners and provide affordable housing to the tenants

Cloud Network Platform Development

Samsung Electronics (HQ), South Korea

Summer Internship (Cloud Software R&D)

May 2019 - July 2019

- Developed a Generic Toolkit for Profiling, Debugging, Automation and Monitoring Health of any Kubernetes Cluster
- Improved the available Samsung Cloud Platform (SCP) Toolkits for incorporating various CNIs (eg. OpenStack Kuryr)

DashCam Car Accident Prediction

ITRI, Hsinchu, Taiwan

Summer Internship (AI/ML Research Project)

May 2018 - July 2018

- Worked with OpenCV, scikit-learn, YOLO, Tensorflow etc. to train car accidents prediction model for self-driving cars

TECHNICAL SKILLS

- **AI & Machine Learning:** PyTorch, TensorFlow, Deep Learning, Computer Vision (OpenCV, YOLO), Reinforcement Learning, scikit-learn
- **Cloud Native & Platforms:** Kubernetes, Docker, Helm, Ansible, Multus, Cilium, DPDK, SR-IOV, OpenStack, KVM, QEMU, Hadoop, Spark, CUDA, OpenMP, MPI
- **Programming Languages:** Python, Go, C, C++, Java, C#, JavaScript, SQL, Bash, VHDL, OCaml, Scala, Rust, LaTeX
- **Web & Apps:** Django, NodeJS, React, Angular, Android Studio, React-Native, Flutter, Git
- **Languages Spoken:** English, Hindi, Korean (TOPIK Level 5)

LICENSES & CERTIFICATIONS

- **Certified Kubernetes Administrator (CKA)** — Cloud Native Computing Foundation (CNCF)
- **Linux Foundation Certified System Administrator (LFCS)** — The Linux Foundation
- **Cloud Specialist** — Samsung Electronics internal certification for cloud platform & infrastructure specialists
- **TOPIK Level 5** (advanced Korean proficiency) · **KIIP Level 6** · **KINAT** — Korean language & integration

LEADERSHIP & COMMUNITY

- **President of IIT Alumni Association of Korea** (IITAAC: iitalumkorea.org) (2023 - Present); **Vice President, Public Relations** (2021 - 2023)
- **Speaker** at the **India-Korea Forum on AI 2026**, Ewha Womans University, Seoul (July 2026)
- **Host** of the Fun Cinema & VFX Booth at **India Day Seoul 2026**, Embassy of India (May 2026)
- **Executive Secretary, Indian Students and Researchers in Korea (ISRK)**; **Core Team Member, Indians in Korea (IIK)** (2025 - Present)
- **Microsoft Student Partner India**, Delhi-NCR Region (2017 - 2019)
- **Overall Coordinator** of Nokia Data Science Hackathon; **Executive & Developer**, DevClub IITD (2017 - 2018)
- **Technical Secretary cum Coordinator** — BHM, BSA, BRCA, AAIP, SAC and **Tryst**, IIT Delhi (2017 - 2019)

MAJOR PROJECTS

Automotive Security System

B.Tech Project, IIT Delhi (2019 - 2020)

- Studied the architecture of Controller Area Network (CAN) bus and Plug-in Hybrid Electric Vehicle (PHEV) Technology
- Devised an improved architecture which tackles the existing security vulnerability without hampering the performance

Game-Bots using ML and AI

AI & ML Course Projects, IIT Delhi (2018 - 2019)

Built Robots for numerous games (eg. Yinsh, Arkanoid) using Q-learning, CNN and various Deep Learning Models

Nokia ML Hackathon Platform

Industry Project, IIT Delhi (2018)

Developed a platform to host large scale Data Science Hackathons using Hadoop, Docker, JupyterHub, NodeJS, Django

Baadal (IITD Computing Cloud)

Cloud Computing, IIT Delhi (2020)

Implemented Load Balancing Module, On-Demand VM migration system and Resource Monitoring dashboard in Baadal